# Niharika Bhasin

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## Education

**New York University** — **Sep 2024 – May 2026**
*Masters in Computer Science* — *Brooklyn, NY*

**SRM Institute of Science and Technology** — **Sep 2020 – May 2024**
*Bachelors of Technology in Computer Science Engineering* — *Delhi, India*

## Technical Skills

**Languages & Platforms:** Python, C/C++, Java, JavaScript, SQL, Linux, Embedded Platforms
**Systems & Architecture:** Distributed Systems, Microservices, Event-Driven Architecture, Agile Software Development
**Cloud & Infrastructure:** AWS (IAM, Lambda, DynamoDB, EC2, S3), Docker, Kubernetes, Terraform, Jenkins, Git
**Data, Backend & Streaming:** FastAPI, Django, Node.js, REST APIs, WebSockets, PostgreSQL, PostGIS, Redis, Kafka
**Geospatial & Algorithms:** Computational Geometry, Spatial Indexing (R-Tree), Routing Systems, GIS Data, Mapbox

## Experience

**Software Engineering Intern** — **June 2025 – August 2025**
*Toricent Labs* — *New York City, USA*

- Architected a massively scalable **serverless backend on AWS (Lambda, API Gateway, DynamoDB)**, engineering robust software functionality for high-throughput workloads exceeding **1M+ monthly API invocations**.
- Spearheaded a **zero-downtime infrastructure migration** to AWS Cloud, utilizing **Terraform (IaC)** to **slash operational expenditure by 30%** and eliminate vendor lock-in.
- Developed **50+ production-grade microservices** in **Node.js** and **Python**, participating in **code reviews** to ensure quality standards and integrating **OAuth2** authentication for enterprise clients.
- Debugged distributed caching performance issues by implementing **Redis (DAX)**, reducing **P99 API latency by 25%** under peak stress tests.
- Maintained **99.9% system uptime** via event-driven workflows and full-stack observability using **AWS CloudWatch**; collaborated with QA to verify deployments met technical specifications, and authored documentation covering **API contracts**, infrastructure design, and deployment runbooks for cross-team knowledge sharing.

**Software Engineering GA, NYU ECE** — **June 2025 – May 2026**
*New York University* — *Brooklyn, NY*

- Engineered a centralized **Lab Operations Portal** using **FastAPI** and **React**, replacing fragmented manual logs for **150+ researchers** and automating equipment scheduling with real-time inventory tracking.
- Architected high-throughput **RESTful APIs** and debugged search bottlenecks via **PostgreSQL** indexing and query optimization, reducing **P99 latency by 45%** during peak academic usage.
- Designed and deployed automated **ETL pipelines** using **Python** and Pandas to synchronize database records, eliminating **100% of redundant workflows** and cutting administrative processing time by **35%**.
- Hardened application security by implementing granular **RBAC** and custom validation middleware, maintaining **99.99% data integrity**; authored technical documentation to support software design requirements.

## Projects

**SafeZone** — *Geospatial Navigation & Routing Engine (Django REST, PostGIS, Mapbox)*

- Architected a containerized full-stack geospatial navigation system using **PostgreSQL/PostGIS** for high-performance spatial indexing; debugged routing calculation overhead, achieving a **40% reduction** via Mapbox engine optimization.
- Secured application access using **AWS IAM** via **AWS Cognito RBAC**; collaborated with QA to implement **Playwright E2E testing** protocols, verifying applications meet user needs and technical specifications.

**CodeSync** — *Real-Time Distributed System & Stream Processing (WebSockets, AWS, Docker)*

- Built a low-latency real-time code collaboration engine using **WebSockets** and **AWS SQS**, achieving **sub-50ms sync delay** across concurrent users to optimize cross-functional teamwork.
- Deployed **multi-tenant Docker sandboxes** on isolated **EC2** nodes via automated **CI/CD pipelines**; authored technical documentation covering system design, API specifications, and deployment procedures.

**Ticket Finder** — *High-Concurrency Distributed Platform (FastAPI, Redis, PostgreSQL)*

- Designed a **high-concurrency distributed system** handling **10k+ concurrent requests** within an Agile software development workflow, ensuring application reliability under heavy stress loads.
- Implemented **continuous testing protocols** to identify bugs and race conditions; debugged software layers via **dual-layer concurrency control**, enhancing performance and overall user experience.